CSD 206 – Computer Organisation

Assignment 1

Questions

Q1. Discuss Moore's Law as explained in the paper, "Moore's Law: Past, Present and Future."

Q2. Is Moore's Law still valid? Support your answer with a suitable explanation.

(Answer both questions in about 350-400 words)

Answers

A1:

Moore's Law was given by Gordon E. Moore, the Director of the R&D Department at Fairchild Semiconductor. He also went on to co-founder of the biggest companies of today, Intel. In 1965, he was asked to predict what would happen in 10 years in the semiconductor industry, and he gave an astonishing prediction and said that it would be possible to fit 65,000 components on a single silicon chip of 6 sq. cm, whereas at that time, only 64 could be accommodated.

Moore's prediction was based on his theory that the number of components (transistors) on a chip would double every year. After 10 years, his predictions were found to be on target and he also gave the explanation for the exponential increase in densities. The industry had gained the ability to make bigger chips with fewer defects and without sacrificing yields, with the use of state-of-the-art technology. Another reason was the finer rendering of images and line widths. An interesting reason for the boom was what Moore called "circuit and device cleverness", where manufacturers tried to squeeze as many components as they could on the wafer.

In 1975, Moore declared that there was no possibility of further increasing the number of components by being clever, and only bigger dice and finer dimensions could be the way forward. On the same note, he modified his law to one in which the density doubled every 18 months. The predictions were once again spot on, and the modified law has held beautifully from 1975 right until 2012, after which saturation has started to take place and Brian Krzanich, former CEO of Intel stated that the duration for doubling of transistors was now closer to 2.5 years.

Moore's Law has been termed as a 'self-fulfilling prophecy' by experts, because everyone in the industry recognizes that if they don't stay on the curve, they will fall behind. So it sort of drives itself. Some of the major developmental factors that helped sustain Moore's Law were the invention of Integrated Circuits (IC), planar transistors, the CMOS Process, DRAM technology and photolithography technology to name the most significant ones.

Now, it is important to note that Moore's Law is not actually a 'law', it's more of an observation and projection of a historical trend, and because of that very reason, it cannot be extrapolated till infinity and expected to hold forever. Moore's Law also had an interesting but unintended effect, which came to be known as Moore's Second Law. R&D, manufacturing and test costs have constantly been on the rise, and the second law therefore states that the capital cost of a semiconductor lab also increases exponentially over time.

A2:

Moore's Law was never actually a 'law', but a simple extrapolation with some observations that led to a prediction. Real circumstances, performance and development have validated it. For a long time now, predictions of the end of the law have been made and have never been successful. However, they continue to be made, and the voices are now bigger and louder than ever before.

Moore's Law stated that the number of components on a silicon chip would increase by a factor of two every year. He later modified his law to state 18 months as the doubling period. What is essential to understand though is that at some point in time, this technology, like all others, will stop growing exponentially and enter the realm of diminishing marginal returns, which means that the output for every additional unit of capital will start to reduce. And this is already quite evident, and is in fact stated by Moore's Second Law, an example of which could be the cost of a new fabrication plant in the US. It was $14 million in 1966 and rose to $1500 million in 1995! These increases can be tolerated as long as the chips improve faster, however that is not the case anymore.

Experts like Brian Krzanich, former CEO of Intel have also stated that the period for doubling of components is now closer to 2.5 years and hence we see that it is not possible to sustain such a sharp exponential growth on a long term basis, especially when the size of the chip is obviously limited and there is only a certain extent up to which transistors can be crammed onto it and there is not much possibility of further increasing the number of components by being clever. An economic threat to Moore's Law is the possibility that transistors become so cheap that there will be no profit in making them cheaper.

One might wonder why the chips cannot be made smaller, and the reasons for that are very complex. Apart from not being very cost-effective, the gate insulator thickness could not have been reduced any further as that would have led to increases in gate leakage current, which would be very undesirable. Further reduction in operation voltage would have caused increased leakage current in the 'OFF' state or very low channel current in the 'ON' state.

So, in my opinion, it is either time to declare Moore's Law no longer valid or to update it yet again, because it is simply not possible to continue on the exponential journey for ever and the industry is sure to witness a period of saturation where eventually the number of transistors on a chip will plateau out and remain stagnant until and unless new breakthroughs are made and even if one were able to manage to do so, the cost of the same would be far too high.